



Exploring the CPython JIT

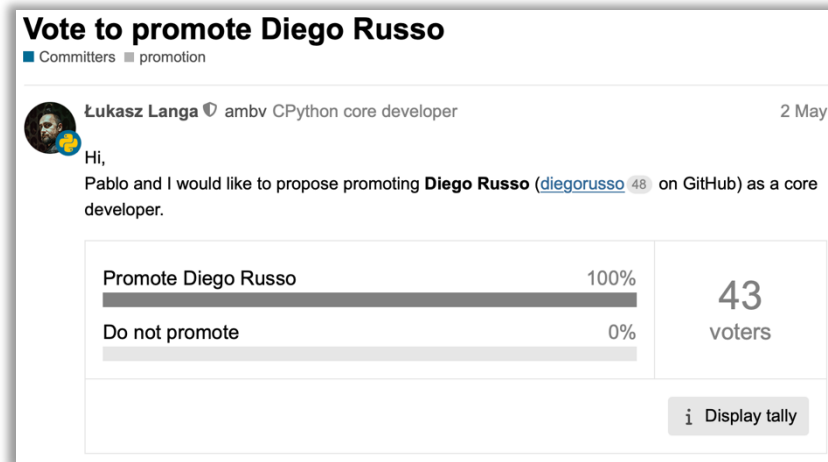
EuroPython 2025

Diego Russo
2025-07-16

Diego Russo

<https://github.com/diegorusso>

- Using Python for almost 20 years
- 14 years at Arm Ltd
 - Principal Software Engineer in the Runtimes team
 - Python Guild with 1500 members
- Contributing to CPython and its ecosystem for 2 years
- Contributing to the new experimental CPython JIT for over a year
- Python Core Developer since May 2025 🎉



What's a JIT compiler?

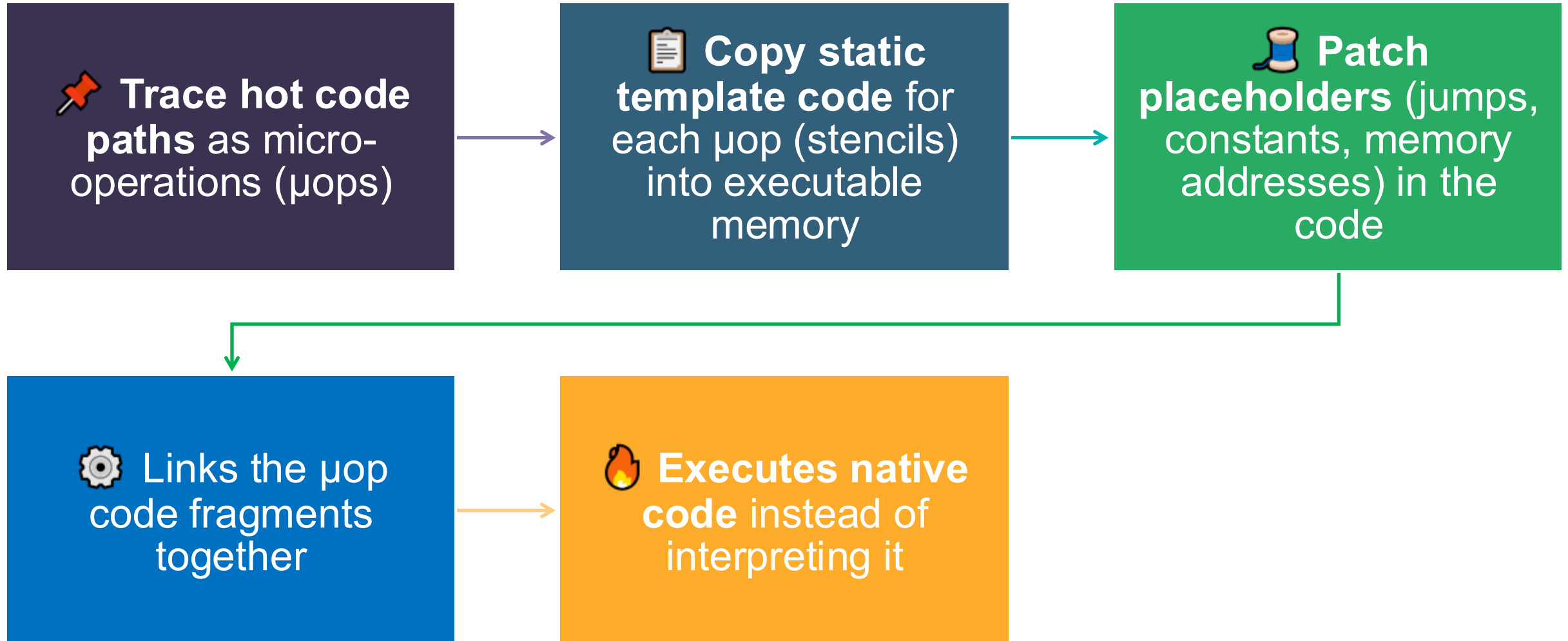
Just-In-Time compiler

A JIT compiler is a component of a runtime system that **dynamically translates code into machine code while the program is running**, rather than before execution.

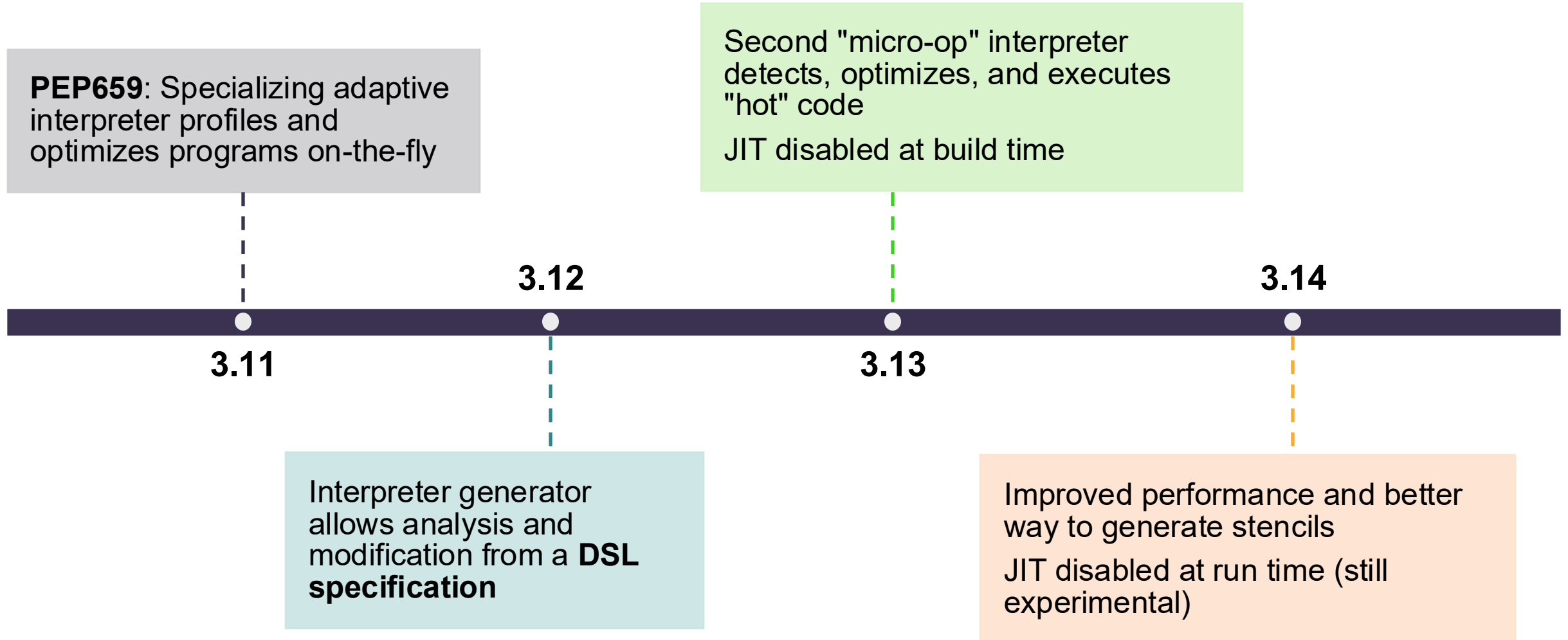
- ✓ Detects and compiles hot code paths at runtime
- ✓ Uses runtime profiling information to optimize code
- ✓ Creates traces of frequently executed operations
- ✓ Eliminates interpreter overhead for performance gains

What's a JIT compiler?

Copy-and-patch



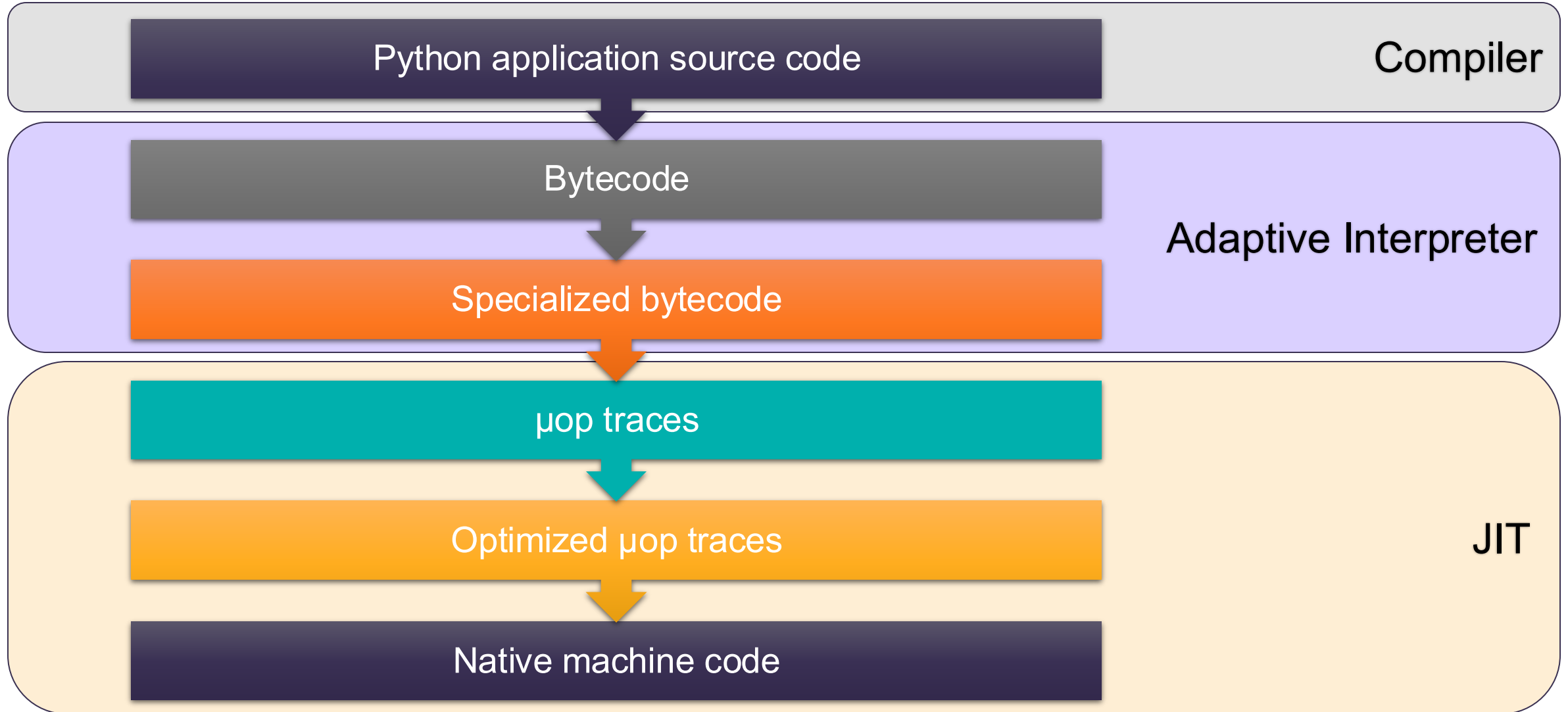
How did we get here?



The CPython JIT in 3.14

The CPython JIT in 3.14

The journey: from source code to machine code.



The CPython JIT in 3.14

Python application source code

```
def sum_squares(n):  
    total = 0  
    for i in range(n):  
        total += i * i  
    return total
```

The CPython JIT in 3.14

Bytecode

```
def sum_squares(n):  
    total = 0  
    for i in range(n):  
        total += i * i  
    return total
```

FOR_ITER	for in :
STORE_FAST	i
LOAD_FAST_LOAD_FAST	total, i
LOAD_FAST	i
BINARY_OP	*
BINARY_OP	+=
STORE_FAST	total
JUMP_BACKWARD	

The CPython JIT in 3.14

Specialized bytecode

```
def sum_squares(n):  
    total = 0  
    for i in range(n):  
        total += i * i  
    return total
```

FOR_ITER	for in :
STORE_FAST	i
LOAD_FAST_LOAD_FAST	total, i
LOAD_FAST	i
BINARY_OP	*
BINARY_OP	+=
STORE_FAST	total
JUMP_BACKWARD	

The CPython JIT in 3.14

Specialized bytecode

```
def sum_squares(n):  
    total = 0  
    for i in range(n):  
        total += i * i  
    return total
```

FOR_ITER_RANGE	for in :
STORE_FAST	i
LOAD_FAST_LOAD_FAST	total, i
LOAD_FAST	i
BINARY_OP_MULTIPLY_INT	*
BINARY_OP_ADD_INT	+=
STORE_FAST	total
JUMP_BACKWARD_JIT	

The CPython JIT in 3.14

μop traces

FOR_ITER_RANGE
STORE_FAST
LOAD_FAST_LOAD_FAST
LOAD_FAST
BINARY_OP_MULTIPLY_INT
BINARY_OP_ADD_INT
STORE_FAST
JUMP_BACKWARD_JIT

The CPython JIT in 3.14

μop traces

_CHECK_VALIDITY

_SET_IP

_ITER_CHECK_RANGE

_GUARD_NOT_EXHAUSTED_RANGE

_ITER_NEXT_RANGE

STORE_FAST

LOAD_FAST_LOAD_FAST

LOAD_FAST

BINARY_OP_MULTIPLY_INT

BINARY_OP_ADD_INT

STORE_FAST

JUMP_BACKWARD_JIT

The CPython JIT in 3.14

μop traces

_CHECK_VALIDITY

_SET_IP

_ITER_CHECK_RANGE

_GUARD_NOT_EXHAUSTED_RANGE

_ITER_NEXT_RANGE

_CHECK_VALIDITY

_SET_IP

_STORE_FAST

LOAD_FAST_LOAD_FAST

LOAD_FAST

BINARY_OP_MULTIPLY_INT

BINARY_OP_ADD_INT

STORE_FAST

JUMP_BACKWARD_JIT

The CPython JIT in 3.14

μop traces

_CHECK_VALIDITY

_SET_IP

_ITER_CHECK_RANGE

_GUARD_NOT_EXHAUSTED_RANGE

_ITER_NEXT_RANGE

_CHECK_VALIDITY

_SET_IP

_STORE_FAST

_CHECK_VALIDITY

_SET_IP

_LOAD_FAST

_LOAD_FAST

LOAD_FAST

BINARY_OP_MULTIPLY_INT

BINARY_OP_ADD_INT

STORE_FAST

JUMP_BACKWARD_JIT

The CPython JIT in 3.14

μop traces

_CHECK_VALIDITY

_SET_IP

_ITER_CHECK_RANGE

_GUARD_NOT_EXHAUSTED_RANGE

_ITER_NEXT_RANGE

_CHECK_VALIDITY

_SET_IP

_STORE_FAST

_CHECK_VALIDITY

_SET_IP

_LOAD_FAST

_LOAD_FAST

_CHECK_VALIDITY

_SET_IP

_LOAD_FAST

BINARY_OP_MULTIPLY_INT

BINARY_OP_ADD_INT

STORE_FAST

JUMP_BACKWARD_JIT

The CPython JIT in 3.14

μop traces

_CHECK_VALIDITY
_SET_IP
_ITER_CHECK_RANGE
_GUARD_NOT_EXHAUSTED_RANGE
_ITER_NEXT_RANGE
_CHECK_VALIDITY
_SET_IP
_STORE_FAST
_CHECK_VALIDITY
_SET_IP
_LOAD_FAST
_LOAD_FAST
_CHECK_VALIDITY
_SET_IP
_LOAD_FAST

_CHECK_VALIDITY
_SET_IP
_GUARD_TOS_INT
_GUARD_NOS_INT
_BINARY_OP_MULTIPLY_INT

BINARY_OP_ADD_INT
STORE_FAST
JUMP_BACKWARD_JIT

The CPython JIT in 3.14

μop traces

_CHECK_VALIDITY
_SET_IP
_ITER_CHECK_RANGE
_GUARD_NOT_EXHAUSTED_RANGE
_ITER_NEXT_RANGE
_CHECK_VALIDITY
_SET_IP
_STORE_FAST
_CHECK_VALIDITY
_SET_IP
_LOAD_FAST
_LOAD_FAST
_CHECK_VALIDITY
_SET_IP
_LOAD_FAST

_CHECK_VALIDITY
_SET_IP
_GUARD_TOS_INT
_GUARD_NOS_INT
_BINARY_OP_MULTIPLY_INT
_CHECK_VALIDITY
_SET_IP
_GUARD_TOS_INT
_GUARD_NOS_INT
_BINARY_OP_ADD_INT

STORE_FAST
JUMP_BACKWARD_JIT

The CPython JIT in 3.14

μop traces

_CHECK_VALIDITY
_SET_IP
_ITER_CHECK_RANGE
_GUARD_NOT_EXHAUSTED_RANGE
_ITER_NEXT_RANGE
_CHECK_VALIDITY
_SET_IP
_STORE_FAST
_CHECK_VALIDITY
_SET_IP
_LOAD_FAST
_LOAD_FAST
_CHECK_VALIDITY
_SET_IP
_LOAD_FAST

_CHECK_VALIDITY
_SET_IP
_GUARD_TOS_INT
_GUARD_NOS_INT
_BINARY_OP_MULTIPLY_INT
_CHECK_VALIDITY
_SET_IP
_GUARD_TOS_INT
_GUARD_NOS_INT
_BINARY_OP_ADD_INT
_CHECK_VALIDITY
_SET_IP
_STORE_FAST

JUMP_BACKWARD_JIT

The CPython JIT in 3.14

μop traces

_CHECK_VALIDITY
_SET_IP
_ITER_CHECK_RANGE
_GUARD_NOT_EXHAUSTED_RANGE
_ITER_NEXT_RANGE
_CHECK_VALIDITY
_SET_IP
_STORE_FAST
_CHECK_VALIDITY
_SET_IP
_LOAD_FAST
_LOAD_FAST
_CHECK_VALIDITY
_SET_IP
_LOAD_FAST

_CHECK_VALIDITY
_SET_IP
_GUARD_TOS_INT
_GUARD_NOS_INT
_BINARY_OP_MULTIPLY_INT
_CHECK_VALIDITY
_SET_IP
_GUARD_TOS_INT
_GUARD_NOS_INT
_BINARY_OP_ADD_INT
_CHECK_VALIDITY
_SET_IP
_STORE_FAST
_CHECK_VALIDITY
_SET_IP
_CHECK_PERIODIC
_JUMP_TO_TOP

The CPython JIT in 3.14

Optimized μ op traces

_CHECK_VALIDITY
_SET_IP
_ITER_CHECK_RANGE
_GUARD_NOT_EXHAUSTED_RANGE
_ITER_NEXT_RANGE
_CHECK_VALIDITY
_SET_IP
_STORE_FAST_2
_CHECK_VALIDITY
_SET_IP
_LOAD_FAST_1
_LOAD_FAST_2
_CHECK_VALIDITY
_SET_IP
_LOAD_FAST_2

_CHECK_VALIDITY
_SET_IP
_GUARD_TOS_INT
_GUARD_NOS_INT
_BINARY_OP_MULTIPLY_INT
_CHECK_VALIDITY
_SET_IP
_GUARD_TOS_INT
_GUARD_NOS_INT
_BINARY_OP_ADD_INT
_CHECK_VALIDITY
_SET_IP
_STORE_FAST_1
_CHECK_VALIDITY
_SET_IP
_CHECK_PERIODIC
_JUMP_TO_TOP

The CPython JIT in 3.14

Optimized μ op traces

```
__CHECK_VALIDITY  
__SET_IP  
__ITER_CHECK_RANGE  
__GUARD_NOT_EXHAUSTED_RANGE  
__ITER_NEXT_RANGE  
__CHECK_VALIDITY  
__SET_IP  
__STORE_FAST_2  
__CHECK_VALIDITY  
__SET_IP  
__LOAD_FAST_1  
__LOAD_FAST_2  
__CHECK_VALIDITY  
__SET_IP  
__LOAD_FAST_2
```

```
__CHECK_VALIDITY  
__SET_IP  
__GUARD_TOS_INT  
__GUARD_NOS_INT  
__BINARY_OP_MULTIPLY_INT  
__CHECK_VALIDITY  
__SET_IP  
__GUARD_TOS_INT  
__GUARD_NOS_INT  
__BINARY_OP_ADD_INT  
__CHECK_VALIDITY  
__SET_IP  
__STORE_FAST_1  
__CHECK_VALIDITY  
__SET_IP  
__CHECK_PERIODIC  
__JUMP_TO_TOP
```

The CPython JIT in 3.14

Optimized μ op traces

_ITER_CHECK_RANGE

_GUARD_NOT_EXHAUSTED_RANGE

_ITER_NEXT_RANGE

_SET_IP

_STORE_FAST_2

_CHECK_VALIDITY

_LOAD_FAST_1

_LOAD_FAST_2

_LOAD_FAST_2

_SET_IP

_BINARY_OP_MULTIPLY_INT

_CHECK_VALIDITY

_SET_IP

_GUARD_NOS_INT

_BINARY_OP_ADD_INT

_STORE_FAST_1

_CHECK_PERIODIC

_JUMP_TO_TOP

The CPython JIT in 3.14

Optimized μ op traces

_ITER_CHECK_RANGE
_GUARD_NOT_EXHAUSTED_RANGE
_ITER_NEXT_RANGE
_SET_IP
_STORE_FAST_2
_CHECK_VALIDITY
_LOAD_FAST_1
_LOAD_FAST_2
_LOAD_FAST_2
_SET_IP
_BINARY_OP_MULTIPLY_INT
_CHECK_VALIDITY
_SET_IP
_GUARD_NOS_INT
_BINARY_OP_ADD_INT
_STORE_FAST_1
_CHECK_PERIODIC
_JUMP_TO_TOP

The CPython JIT in 3.14

Native machine code

_ITER_CHECK_RANGE
_GUARD_NOT_EXHAUSTED_RANGE
_ITER_NEXT_RANGE
_SET_IP
_STORE_FAST_2
_CHECK_VALIDITY
_LOAD_FAST_1
_LOAD_FAST_2
_LOAD_FAST_2
_SET_IP
_BINARY_OP_MULTIPLY_INT
_CHECK_VALIDITY
_SET_IP
_GUARD_NOS_INT
_BINARY_OP_ADD_INT
_STORE_FAST_1
_CHECK_PERIODIC
_JUMP_TO_TOP

The CPython JIT in 3.14

Native machine code

```
a8 82 5f f8 e9 03 28 2a 3f 05 40 f2 20 01 00 54 08 f9 7f 92 09 00 00 d0 08 05 40 f9 29 ed 46 f9 1f 01 09 eb 40 00 00 54 66 06 00 14 0d 00 00 14 fd 7b bf a9 08 00 00 d0 00 00 00 d0 00 80 35
91 08 f1 46 f9 01 00 00 d0 21 f4 35 91 03 00 00 d0 63 80 36 91 22 3f 80 52 fd 03 00 91 00 01 3f d6
```

_GUARD_NOT_EXHAUSTED_RANGE

_ITER_NEXT_RANGE

_SET_IP

_STORE_FAST_2

_CHECK_VALIDITY

_LOAD_FAST_1

_LOAD_FAST_2

_LOAD_FAST_2

_SET_IP

_BINARY_OP_MULTIPLY_INT

_CHECK_VALIDITY

_SET_IP

_GUARD_NOS_INT

_BINARY_OP_ADD_INT

_STORE_FAST_1

_CHECK_PERIODIC

_JUMP_TO_TOP

The CPython JIT in 3.14

Native machine code

```
a8 82 5f f8 e9 03 28 2a 3f 05 40 f2 20 01 00 54 08 f9 7f 92 09 00 00 d0 08 05 40 f9 29 ed 46 f9 1f 01 09 eb 40 00 00 54 66 06 00 14 0d 00 00 14 fd 7b bf a9 08 00 00 d0 00 00 00 d0 00 80 35
91 08 f1 46 f9 01 00 00 d0 21 f4 35 91 03 00 00 d0 63 80 36 91 22 3f 80 52 fd 03 00 91 00 01 3f d6 fd 7b bf a9 a8 82 5f f8 fd 03 00 91 e9 03 28 2a 3f 05 40 f2 c0 01 00 54 08 f9 7f 92 0a 00
00 d0 09 05 40 f9 4a 75 47 f9 3f 01 0a eb 41 02 00 54 08 11 40 f9 1f 01 00 f1 6d 00 00 54 fd 7b c1 a8 18 00 00 14 fd 7b c1 a8 0c 07 00 14 08 00 00 d0 00 00 00 d0 00 d4 39 91 08 79 47 f9 01
00 00 d0 21 48 3a 91 03 00 00 d0 63 d4 3a 91 22 3f 80 52 00 01 3f d6 08 00 00 d0 00 00 00 d0 00 a0 37 91 08 79 47 f9 01 00 00 d0 21 20 38 91 03 00 00 d0 63 94 38 91 42 0f 80 52 00 01 3f d6
1f 20 03 d5
```

_ITER_NEXT_RANGE

_SET_IP

_STORE_FAST_2

_CHECK_VALIDITY

_LOAD_FAST_1

_LOAD_FAST_2

_LOAD_FAST_2

_SET_IP

_BINARY_OP_MULTIPLY_INT

_CHECK_VALIDITY

_SET_IP

_GUARD_NOS_INT

_BINARY_OP_ADD_INT

_STORE_FAST_1

_CHECK_PERIODIC

_JUMP_TO_TOP

The CPython JIT in 3.14

Native machine code

`_SET_IP`
`_STORE_FAST_2`
`_CHECK_VALIDITY`
`_LOAD_FAST_1`
`_LOAD_FAST_2`
`_LOAD_FAST_2`
`_SET_IP`
`_BINARY_OP_MULTIPLY_INT`
`_CHECK_VALIDITY`
`_SET_IP`
`_GUARD_NOS_INT`
`_BINARY_OP_ADD_INT`
`_STORE_FAST_1`
`_CHECK_PERIODIC`
`_JUMP_TO_TOP`

```
a8 82 5f f8 e9 03 28 2a 3f 05 40 f2 20 01 00 54 08 f9 7f 92 09 00 00 d0 08 05 40 f9 29 ed 46 f9 1f 01 09 eb 40 00 00 54 66 06 00 14 0d 00 00 14 fd 7b bf a9 08 00 00 d0 00 00 00 d0 00 80 35
91 08 f1 46 f9 01 00 00 d0 21 f4 35 91 03 00 00 d0 63 80 36 91 22 3f 80 52 fd 03 00 91 00 01 3f d6 fd 7b bf a9 a8 82 5f f8 fd 03 00 91 e9 03 28 2a 3f 05 40 f2 c0 01 00 54 08 f9 7f 92 0a 00
00 d0 09 05 40 f9 4a 75 47 f9 3f 01 0a eb 41 02 00 54 08 11 40 f9 1f 01 00 f1 6d 00 00 54 fd 7b c1 a8 18 00 00 14 fd 7b c1 a8 0c 07 00 14 08 00 00 d0 00 d4 39 91 08 79 47 f9 01
00 00 d0 21 48 3a 91 03 00 00 d0 63 d4 3a 91 22 3f 80 52 00 01 3f d6 08 00 00 d0 00 a0 37 91 08 79 47 f9 01 00 00 d0 21 20 38 91 03 00 00 d0 63 94 38 91 42 0f 80 52 00 01 3f d6
1f 20 03 d5 fd 7b bf a9 a8 82 5f f8 fd 03 00 91 e9 03 28 2a 3f 05 40 f2 e0 06 00 54 08 f9 7f 92 0a 00 00 f0 09 05 40 f9 4a 79 40 f9 3f 01 0a eb 61 07 00 54 09 11 40 f9 3f 01 00 f1 4d 08 00
91 62 0f 80 52 00 01 3f d6 08 00 00 f0 00 00 00 d0 00 ec 3d 91 08 85 40 f9 01 00 00 d0 21 60 3c 91 03 00 00 d0 63 ac 3c 91 e2 0f 80 52 00 01 3f d6 08 00 00 f0 00 00 d0 00 18 3e 91 08 85
40 f9 01 00 00 d0 21 60 3c 91 03 00 00 d0 63 ac 3c 91 42 11 80 52 00 01 3f d6 08 00 00 f0 00 00 00 00 f0 00 dc 00 91 08 85 40 f9 01 00 00 d0 21 e4 3e 91 03 00 00 f0 63 3c 00 91 62 3b 80 52 00
01 3f d6 08 00 00 f0 00 00 00 f0 00 48 01 91 08 85 40 f9 01 00 00 f0 21 d8 01 91 03 00 00 f0 63 70 02 91 a2 02 80 52 00 01 3f d6 08 00 00 f0 00 00 00 f0 00 48 03 91 08 85 40 f9 01 00 00 f0
21 d8 01 91 03 00 00 f0 63 70 02 91 e2 02 80 52 00 01 3f d6
```

The CPython JIT in 3.14

Native machine code

_STORE_FAST_2

_CHECK_VALIDITY

_LOAD_FAST_1

_LOAD_FAST_2

_LOAD_FAST_2

_SET_IP

_BINARY_OP_MULTIPLY_INT

_CHECK_VALIDITY

_SET_IP

_GUARD_NOS_INT

_BINARY_OP_ADD_INT

_STORE_FAST_1

_CHECK_PERIODIC

_JUMP_TO_TOP

```
a8 82 5f f8 e9 03 28 2a 3f 05 40 f2 20 01 00 54 08 f9 7f 92 09 00 00 d0 08 05 40 f9 29 ed 46 f9 1f 01 09 eb 40 00 00 54 66 06 00 14 0d 00 00 14 fd 7b bf a9 08 00 00 d0 00 00 00 d0 00 80 35
91 08 f1 46 f9 01 00 00 d0 21 f4 35 91 03 00 00 d0 63 80 36 91 22 3f 80 52 fd 03 00 91 00 01 3f d6 fd 7b bf a9 a8 82 5f f8 fd 03 00 91 e9 03 28 2a 3f 05 40 f2 c0 01 00 54 08 f9 7f 92 0a 00
00 d0 09 05 40 f9 4a 75 47 f9 3f 01 0a eb 41 02 00 54 08 11 40 f9 1f 01 00 f1 6d 00 00 54 fd 7b c1 a8 18 00 00 14 fd 7b c1 a8 0c 07 00 14 08 00 00 d0 00 44 39 91 08 79 47 f9 01
00 00 d0 21 48 3a 91 03 00 00 d0 63 d4 3a 91 22 3f 80 52 00 01 3f d6 08 00 00 d0 00 a0 37 91 08 79 47 f9 01 00 00 d0 21 20 38 91 03 00 00 d0 63 94 38 91 42 0f 80 52 00 01 3f d6
1f 20 03 d5 fd 7b bf a9 a8 82 5f f8 fd 03 00 91 e9 03 28 2a 3f 05 40 f2 e0 06 00 54 08 f9 7f 92 0a 00 00 f0 09 05 40 f9 4a 79 40 f9 3f 01 0a eb 61 07 00 54 09 11 40 f9 3f 01 00 f1 4d 08 00
91 62 0f 80 52 00 01 3f d6 08 00 00 f0 00 00 00 d0 00 00 ec 3d 91 08 85 40 f9 01 00 00 d0 21 60 3c 91 03 00 00 d0 63 ac 3c 91 e2 0f 80 52 00 01 3f d6 08 00 00 f0 00 00 d0 00 18 3e 91 08 85
40 f9 01 00 00 d0 21 60 3c 91 03 00 00 d0 63 ac 3c 91 42 11 80 52 00 01 3f d6 08 00 00 f0 00 00 00 00 f0 00 dc 00 91 08 85 40 f9 01 00 00 d0 21 e4 3e 91 03 00 00 f0 63 3c 00 91 62 3b 80 52 00
01 3f d6 08 00 00 f0 00 00 00 f0 00 48 01 91 08 85 40 f9 01 00 00 f0 21 d8 01 91 03 00 00 f0 63 70 02 91 a2 02 80 52 00 01 3f d6 08 00 00 f0 00 00 00 f0 00 48 03 91 08 85 40 f9 01 00 00 f0
21 d8 01 91 03 00 00 f0 63 70 02 91 e2 02 80 52 00 01 3f d6 88 62 01 58 1f 20 03 d5 88 1e 00 f9 1f 20 03 d5
```

The CPython JIT in 3.14

Native machine code

_CHECK_VALIDITY

_LOAD_FAST_1

_LOAD_FAST_2

_LOAD_FAST_2

_SET_IP

_BINARY_OP_MULTIPLY_INT

_CHECK_VALIDITY

_SET_IP

_GUARD_NOS_INT

_BINARY_OP_ADD_INT

_STORE_FAST_1

_CHECK_PERIODIC

_JUMP_TO_TOP

```
a8 82 5f f8 e9 03 28 2a 3f 05 40 f2 20 01 00 54 08 f9 7f 92 09 00 00 d0 08 05 40 f9 29 ed 46 f9 1f 01 09 eb 40 00 00 54 66 06 00 14 0d 00 00 14 fd 7b bf a9 08 00 00 d0 00 00 00 d0 00 80 35
91 08 f1 46 f9 01 00 00 d0 21 f4 35 91 03 00 00 d0 63 80 36 91 22 3f 80 52 fd 03 00 91 00 01 3f d6 fd 7b bf a9 a8 82 5f f8 fd 03 00 91 e9 03 28 2a 3f 05 40 f2 c0 01 00 54 08 f9 7f 92 0a 00
00 d0 09 05 40 f9 4a 75 47 f9 3f 01 0a eb 41 02 00 54 08 11 40 f9 1f 01 00 f1 6d 00 00 54 fd 7b c1 a8 18 00 00 14 fd 7b c1 a8 0c 07 00 14 08 00 00 d0 00 00 00 d0 00 d4 39 91 08 79 47 f9 01
00 00 d0 21 48 3a 91 03 00 00 d0 63 d4 3a 91 22 3f 80 52 00 01 3f d6 08 00 00 d0 00 00 00 d0 00 00 00 a0 37 91 08 79 47 f9 01 00 00 d0 21 20 38 91 03 00 00 d0 63 94 38 91 42 0f 80 52 00 01 3f d6
1f 20 03 d5 fd 7b bf a9 a8 82 5f f8 fd 03 00 91 e9 03 28 2a 3f 05 40 f2 e0 06 00 54 08 f9 7f 92 0a 00 00 f0 09 05 40 f9 4a 79 40 f9 3f 01 0a eb 61 07 00 54 09 11 40 f9 3f 01 00 f1 4d 08 00
91 62 0f 80 52 00 01 3f d6 08 00 00 f0 00 00 00 00 d0 00 ec 3d 91 08 85 40 f9 01 00 00 d0 21 60 3c 91 03 00 00 d0 63 ac 3c 91 e2 0f 80 52 00 01 3f d6 08 00 00 f0 00 00 00 d0 00 18 3e 91 08 85
40 f9 01 00 00 d0 21 60 3c 91 03 00 00 d0 63 ac 3c 91 42 11 80 52 00 01 3f d6 08 00 00 f0 00 00 00 00 f0 00 dc 00 91 08 85 40 f9 01 00 00 d0 21 e4 3e 91 03 00 00 f0 63 3c 00 91 62 3b 80 52 00
01 3f d6 08 00 00 f0 00 00 00 00 00 48 01 91 08 85 40 f9 01 00 00 f0 21 d8 01 91 03 00 00 f0 63 70 02 91 e2 02 80 52 00 01 3f d6 88 62 01 58 1f 20 03 d5 88 1e 00 f9 1f 20 03 d5 fd 7b bf a9 08 00 00 f0 fd 03 00 91 08 4d 42 f9 08 3d 00 12 1f 09 00 71 e1 0a 00
21 d8 01 91 03 00 00 f0 63 70 02 91 e2 02 80 52 00 01 3f d6 88 62 01 58 1f 20 03 d5 88 1e 00 f9 1f 20 03 d5 fd 7b bf a9 08 00 00 f0 fd 03 00 91 08 4d 42 f9 08 3d 00 12 1f 09 00 71 e1 0a 00
54 88 02 40 f9 1f 05 00 f1 c0 0b 00 54 e9 03 28 2a 3f 05 40 f2 a0 0c 00 54 08 f9 7f 92 09 00 00 f0 0a 05 40 f9 29 51 42 f9 5f 01 09 eb 21 08 00 54 0b c1 40 39 0c 14 80 52 aa 8e 5f f8 7f 01
0c 6a e0 00 00 54 5f 05 00 f1 a0 00 00 54 8a 00 00 36 4b f9 7f 92 6b 01 40 b9 4b 13 f8 36 8b 2a 41 39 93 32 40 f9 8a 32 00 f9 7f 0d 00 71 a0 01 00 54 0a 05 40 f9 5f 01 09 eb c1 05 00 54 09
49 80 b9 89 0e 09 8b a9 02 09 cb 29 41 01 d1 29 0a 10 b7 29 fd 43 d3 08 41 40 b9 1f 01 09 6b ab 09 00 54 88 22 40 f9 a8 0a 00 b5 95 22 00 f9 b3 0b 00 b4 b3 03 00 37 7f 06 00 f1 80 0c 00 54
68 02 40 b9 08 01 00 35 08 00 00 f0 00 00 00 f0 00 20 0b 91 08 55 42 f9 41 53 80 52 e2 03 13 aa 00 01 3f d6 68 0e 40 79 68 0c 10 37 68 02 40 b9 a8 00 f8 37 a8 76 01 58 1f 20 03 d5 00 01 3f
d6 68 02 40 b9 08 05 00 71 68 02 00 b9 a1 00 00 54 08 00 00 f0 e0 03 13 aa 08 5d 42 f9 00 01 3f d6 95 22 40 f9 d5 0b 00 b4 9f 22 00 f9 fd 7b c1 a8 70 00 00 14 08 00 00 f0 00 00 00 f0 00 48
0a 91 08 61 42 f9 01 00 00 f0 21 d8 08 91 03 00 00 f0 63 70 09 91 e2 02 80 52 00 01 3f d6 08 00 00 f0 00 00 00 f0 00 60 04 91 08 61 42 f9 01 00 00 f0 21 c4 04 91 03 00 00 f0 63 04 05 91 22
0f 80
```

The CPython JIT in 3.14

Native machine code

_LOAD_FAST_1

_LOAD_FAST_2

_LOAD_FAST_2

_SET_IP

_BINARY_OP_MULTIPLY_INT

_CHECK_VALIDITY

_SET_IP

_GUARD_NOS_INT

_BINARY_OP_ADD_INT

_STORE_FAST_1

_CHECK_PERIODIC

_JUMP_TO_TOP

```
a8 82 5f f8 e9 03 28 2a 3f 05 40 f2 20 01 00 54 08 f9 7f 92 09 00 00 d0 08 05 40 f9 29 ed 46 f9 1f 01 09 eb 40 00 00 54 66 06 00 14 0d 00 00 14 fd 7b bf a9 08 00 00 d0 00 00 00 d0 00 80 35
91 08 f1 46 f9 01 00 00 d0 21 f4 35 91 03 00 00 d0 63 80 36 91 22 3f 80 52 fd 03 00 91 00 01 3f d6 fd 7b bf a9 a8 82 5f f8 fd 03 00 91 e9 03 28 2a 3f 05 40 f2 c0 01 00 54 08 f9 7f 92 0a 00
00 d0 09 05 40 f9 4a 75 47 f9 3f 01 0a eb 41 02 00 54 08 11 40 f9 1f 01 00 f1 6d 00 00 54 fd 7b c1 a8 18 00 00 14 fd 7b c1 a8 0c 07 00 14 08 00 00 d0 00 00 00 d0 00 d4 39 91 08 79 47 f9 01
00 00 d0 21 48 3a 91 03 00 00 d0 63 d4 3a 91 22 3f 80 52 00 01 3f d6 08 00 00 d0 00 00 00 d0 00 00 00 a0 37 91 08 79 47 f9 01 00 00 d0 21 20 38 91 03 00 00 d0 63 94 38 91 42 0f 80 52 00 01 3f d6
1f 20 03 d5 fd 7b bf a9 a8 82 5f f8 fd 03 00 91 e9 03 28 2a 3f 05 40 f2 e0 06 00 54 08 f9 7f 92 0a 00 00 f0 09 05 40 f9 4a 79 40 f9 3f 01 0a eb 61 07 00 54 09 11 40 f9 3f 01 00 f1 4d 08 00
91 62 0f 80 52 00 01 3f d6 08 00 00 f0 00 00 00 00 d0 00 ec 3d 91 08 85 40 f9 01 00 00 d0 21 60 3c 91 03 00 00 d0 63 ac 3c 91 e2 0f 80 52 00 01 3f d6 08 00 00 f0 00 00 00 d0 00 18 3e 91 08 85
40 f9 01 00 00 d0 21 60 3c 91 03 00 00 d0 63 ac 3c 91 42 11 80 52 00 01 3f d6 08 00 00 f0 00 00 00 00 f0 00 dc 00 91 08 85 40 f9 01 00 00 d0 21 e4 3e 91 03 00 00 f0 63 3c 00 91 62 3b 80 52 00
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21 d8 01 91 03 00 00 f0 63 70 02 91 e2 02 80 52 00 01 3f d6 88 62 01 58 1f 20 03 d5 88 1e 00 f9 1f 20 03 d5 fd 7b bf a9 08 00 00 f0 fd 03 00 91 08 4d 42 f9 08 3d 00 12 1f 09 00 71 e1 0a 00
54 88 02 40 f9 1f 05 00 f1 c0 0b 00 54 e9 03 28 2a 3f 05 40 f2 a0 0c 00 54 08 f9 7f 92 09 00 00 f0 0a 05 40 f9 29 51 42 f9 5f 01 09 eb 21 08 00 54 0b c1 40 39 0c 14 80 52 aa 8e 5f f8 7f 01
0c 6a e0 00 00 54 5f 05 00 f1 a0 00 00 54 8a 00 00 36 4b f9 7f 92 6b 01 40 b9 4b 13 f8 36 8b 2a 41 39 93 32 40 f9 8a 32 00 f9 7f 0d 00 71 a0 01 00 54 0a 05 40 f9 5f 01 09 eb c1 05 00 54 09
49 80 b9 89 0e 09 8b a9 02 09 cb 29 41 01 d1 29 0a 10 b7 29 fd 43 d3 08 41 40 b9 1f 01 09 6b ab 09 00 54 88 22 40 f9 a8 0a 00 b5 95 22 00 f9 b3 0b 00 b4 b3 03 00 37 7f 06 00 f1 80 0c 00 54
68 02 40 b9 08 01 00 35 08 00 00 f0 00 00 00 f0 00 20 0b 91 08 55 42 f9 41 53 80 52 e2 03 13 aa 00 01 3f d6 68 0e 40 79 68 0c 10 37 68 02 40 b9 a8 00 f8 37 a8 76 01 58 1f 20 03 d5 00 01 3f
d6 68 02 40 b9 08 05 00 71 68 02 00 b9 a1 00 00 54 08 00 00 f0 e0 03 13 aa 08 5d 42 f9 00 01 3f d6 95 22 40 f9 d5 0b 00 b4 9f 22 00 f9 fd 7b c1 a8 70 00 00 14 08 00 00 f0 00 00 00 f0 00 48
0a 91 08 61 42 f9 01 00 00 f0 21 d8 08 91 03 00 00 f0 63 70 09 91 e2 02 80 52 00 01 3f d6 08 00 00 f0 00 00 00 f0 00 60 04 91 08 61 42 f9 01 00 00 f0 21 c4 04 91 03 00 00 f0 63 04 05 91 22
0f 80 88 67 01 58 1f 20 03 d5 08 89 40 39 48 00 00 37 c0 06 00 14 1f 20 03 d5
```

The CPython JIT in 3.14

Native machine code

_SET_IP

_BINARY_OP_MULTIPLY_INT

_CHECK_VALIDITY

_SET_IP

_GUARD_NOS_INT

_BINARY_OP_ADD_INT

_STORE_FAST_1

_CHECK_PERIODIC

_JUMP_TO_TOP

```
a8 82 5f f8 e9 03 28 2a 3f 05 40 f2 20 01 00 54 08 f9 7f 92 09 00 00 d0 08 05 40 f9 29 ed 46 f9 1f 01 09 eb 40 00 00 54 66 06 00 14 0d 00 00 14 fd 7b bf a9 08 00 00 d0 00 00 00 d0 00 80 35
91 08 f1 46 f9 01 00 00 d0 21 f4 35 91 03 00 00 d0 63 80 36 91 22 3f 80 52 fd 03 00 91 00 01 3f d6 fd 7b bf a9 a8 82 5f f8 fd 03 00 91 e9 03 28 2a 3f 05 40 f2 c0 01 00 54 08 f9 7f 92 0a 00
00 d0 09 05 40 f9 4a 75 47 f9 3f 01 0a eb 41 02 00 54 08 11 40 f9 1f 01 00 f1 6d 00 00 54 fd 7b c1 a8 18 00 00 14 fd 7b c1 a8 0c 07 00 14 08 00 00 d0 00 00 00 d0 00 d4 39 91 08 79 47 f9 01
00 00 d0 21 48 3a 91 03 00 00 d0 63 d4 3a 91 22 3f 80 52 00 01 3f d6 08 00 00 d0 00 00 00 d0 a0 37 91 08 79 47 f9 01 00 00 d0 21 20 38 91 03 00 00 d0 63 94 38 91 42 0f 80 52 00 01 3f d6
1f 20 03 d5 fd 7b bf a9 a8 82 5f f8 fd 03 00 91 e9 03 28 2a 3f 05 40 f2 e0 06 00 54 08 f9 7f 92 0a 00 00 f0 09 05 40 f9 4a 79 40 f9 3f 01 0a eb 61 07 00 54 09 11 40 f9 3f 01 00 f1 4d 08 00
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0c 6a e0 00 00 54 5f 05 00 f1 a0 00 00 54 8a 00 00 36 4b f9 7f 92 6b 01 40 b9 4b 13 f8 36 8b 2a 41 39 93 32 40 f9 8a 32 00 f9 7f 0d 00 71 a0 01 00 54 0a 05 40 f9 5f 01 09 eb c1 05 00 54 09
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68 02 40 b9 08 01 00 35 08 00 00 f0 00 00 00 f0 00 20 0b 91 08 55 42 f9 41 53 80 52 e2 03 13 aa 00 01 3f d6 68 0e 40 79 68 0c 10 37 68 02 40 b9 a8 00 f8 37 a8 76 01 58 1f 20 03 d5 00 01 3f
d6 68 02 40 b9 08 05 00 71 68 02 00 b9 a1 00 00 54 08 00 00 f0 e0 03 13 aa 08 5d 42 f9 00 01 3f d6 95 22 40 f9 d5 0b 00 b4 9f 22 00 f9 fd 7b c1 a8 70 00 00 14 08 00 00 f0 00 00 00 f0 00 48
0a 91 08 61 42 f9 01 00 00 f0 21 d8 08 91 03 00 00 f0 63 70 09 91 e2 02 80 52 00 01 3f d6 08 00 00 f0 00 00 00 f0 00 60 04 91 08 61 42 f9 01 00 00 f0 21 c4 04 91 03 00 00 f0 63 04 05 91 22
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80 52 00 01 3f d6 08 00 00 f0 00 00 00 f0 00 cc 15 91 08 55 43 f9 01 00 00 f0 21 a4 13 91 03 00 00 f0 63 fc 13 91 c2 0f 80 52 00 01 3f d6 08 00 00 f0 00 00 00 f0 00 24 16 91 08 55 43 f9 01
00 00 f0 21 b4 16 91 03 00 00 f0 63 4c 17 91 e2 02 80 52 00 01 3f d6 08 00 00 f0 00 00 00 f0 00 88 18 91 08 55 43 f9 01 00 00 f0 21 b4 16 91 03 00 00 f0 63 4c 17 91 e2 02 80 52 00 01 3f
71 01 04 00 54 88 2e 40 f9 1f 05 00 f1 e0 04 00 54 08 01 40 b2 a8 86 00 f8 88 2a 41 39 1f 0d 00 71 c0 02 00 54 88 02 40 f9 1f 05 00 f1 60 06 00 54 e9 03 28 2a 3f 05 40 f2 40 07 00 54 08 f9
7f 92 0a 00 00 f0 09 05 40 f9 4a 51 43 f9 3f 01 0a eb c1 07 00 54 09 49 80 b9 89 0e 09 8b a9 02 09 cb 29 41 01 d1 69 03 10 b7 29 fd 43 d3 08 41 40 b9 1f 01 09 6b eb 02 00 54 fd 7b c1 a8 3d
00 00 14 08 00 00 f0 00 00 00 f0 00 40 13 91 08 55 43 f9 01 00 00 f0 21 a4 13 91 03 00 00 f0 63 fc 13 91 22 0f 80 52 00 01 3f d6 08 00 00 f0 00 00 00 f0 00 3c 15 91 08 55 43 f9 01 00 00 f0
21 a4 13 91 03 00 00 f0 63 fc 13 91 42 0f 80 52 00 01 3f d6 08 00 00 f0 00 cc 15 91 08 55 43 f9 01 00 00 f0 21 a4 13 91 03 00 00 f0 63 fc 13 91 c2 0f 80 52 00 01 3f d6 08 00 00
f0 00 00 00 f0 00 24 16 91 08 55 43 f9 01 00 00 f0 21 b4 16 91 03 00 00 f0 63 4c 17 91 a2 02 80 52 00 01 3f d6 08 00 00 f0 00 00 00 f0 00 88 18 91 08 55 43 f9 01 00 00 f0 21 b4 16 91 03 00
00 f0 63 88 19 91 22 3f 80 52 00 01 3f d6 08 00 00 f0 00 24 18 91 08 55 43 f9 01 00 00 f0 21 b4 16 91 03 00 00 f0 63 4c 17 91 e2 02 80 52 00 01 3f d6
```

The CPython JIT in 3.14

Native machine code

_BINARY_OP_MULTIPLY_INT

_CHECK_VALIDITY

_SET_IP

_GUARD_NOS_INT

_BINARY_OP_ADD_INT

_STORE_FAST_1

_CHECK_PERIODIC

_JUMP_TO_TOP

```
a8 82 5f f8 e9 03 28 2a 3f 05 40 f2 20 01 00 54 08 f9 7f 92 09 00 00 d0 08 05 40 f9 29 ed 46 f9 1f 01 09 eb 40 00 00 54 66 06 00 14 0d 00 00 14 fd 7b bf a9 08 00 00 d0 00 00 00 d0 00 80 35
91 08 f1 46 f9 01 00 00 d0 21 f4 35 91 03 00 00 d0 63 80 36 91 22 3f 80 52 fd 03 00 91 00 01 3f d6 fd 7b bf a9 a8 82 5f f8 fd 03 00 91 e9 03 28 2a 3f 05 40 f2 c0 01 00 54 08 f9 7f 92 0a 00
00 d0 09 05 40 f9 4a 75 47 f9 3f 01 0a eb 41 02 00 54 08 11 40 f9 1f 01 00 f1 6d 00 00 54 fd 7b c1 a8 18 00 00 14 fd 7b c1 a8 0c 07 00 14 08 00 00 d0 00 44 39 91 08 79 47 f9 01
00 00 d0 21 48 3a 91 03 00 00 d0 63 d4 3a 91 22 3f 80 52 00 01 3f d6 08 00 00 d0 00 00 00 d0 a0 37 91 08 79 47 f9 01 00 00 d0 21 20 38 91 03 00 00 d0 63 94 38 91 42 0f 80 52 00 01 3f d6
1f 20 03 d5 fd 7b bf a9 a8 82 5f f8 fd 03 00 91 e9 03 28 2a 3f 05 40 f2 e0 06 00 54 08 f9 7f 92 0a 00 00 f0 09 05 40 f9 4a 79 40 f9 3f 01 0a eb 61 07 00 54 09 11 40 f9 3f 01 00 f1 4d 08 00
91 62 0f 80 52 00 01 3f d6 08 00 00 f0 00 00 00 00 d0 00 ec 3d 91 08 85 40 f9 01 00 00 d0 21 60 3c 91 03 00 00 d0 63 ac 3c 91 e2 0f 80 52 00 01 3f d6 08 00 00 f0 00 00 00 d0 18 3e 91 08 85
40 f9 01 00 00 d0 21 60 3c 91 03 00 00 d0 63 ac 3c 91 42 11 80 52 00 01 3f d6 08 00 00 f0 00 00 00 00 f0 00 dc 00 91 08 85 40 f9 01 00 00 d0 21 e4 3e 91 03 00 00 f0 63 3c 00 91 62 3b 80 52 00
01 3f d6 08 00 00 f0 00 00 00 f0 00 48 01 91 08 85 40 f9 01 00 00 f0 21 d8 01 91 03 00 00 f0 63 70 02 91 e2 02 80 52 00 01 3f d6 88 62 01 58 1f 20 03 d5 88 1e 00 f9 1f 20 03 d5 fd 7b bf a9 08 00 00 f0 fd 03 00 91 08 4d 42 f9 08 3d 00 12 1f 09 00 71 e1 0a 00
21 d8 01 91 03 00 00 f0 63 70 02 91 e2 02 80 52 00 01 3f d6 88 62 01 58 1f 20 03 d5 88 1e 00 f9 1f 20 03 d5 fd 7b bf a9 08 00 00 f0 fd 03 00 91 08 4d 42 f9 08 3d 00 12 1f 09 00 71 e1 0a 00
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0c 6a e0 00 00 54 5f 05 00 f1 a0 00 00 54 8a 00 00 36 4b f9 7f 92 6b 01 40 b9 4b 13 f8 36 8b 2a 41 39 93 32 40 f9 8a 32 00 f9 7f 0d 00 71 a0 01 00 54 0a 05 40 f9 5f 01 09 eb c1 05 00 54 09
49 80 b9 89 0e 09 8b a9 02 09 cb 29 41 01 d1 29 0a 10 b7 29 fd 43 d3 08 41 40 b9 1f 01 09 6b ab 09 00 54 88 22 40 f9 a8 0a 00 b5 95 22 00 f9 b3 0b 00 b4 b3 03 00 37 7f 06 00 f1 80 0c 00 54
68 02 40 b9 08 01 00 35 08 00 00 f0 00 00 00 f0 00 20 0b 91 08 55 42 f9 41 53 80 52 e2 03 13 aa 00 01 3f d6 68 0e 40 79 68 0c 10 37 68 02 40 b9 a8 00 f8 37 a8 76 01 58 1f 20 03 d5 00 01 3f
d6 68 02 40 b9 08 05 00 71 68 02 00 b9 a1 00 00 54 08 00 00 f0 e0 03 13 aa 08 5d 42 f9 00 01 3f d6 95 22 40 f9 d5 0b 00 b4 9f 22 00 f9 fd 7b c1 a8 70 00 00 14 08 00 00 f0 00 00 00 f0 00 48
0a 91 08 61 42 f9 01 00 00 f0 21 d8 08 91 03 00 00 f0 63 70 09 91 e2 02 80 52 00 01 3f d6 08 00 00 f0 00 00 00 f0 00 60 04 91 08 61 42 f9 01 00 00 f0 21 c4 04 91 03 00 00 f0 63 04 05 91 22
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80 52 00 01 3f d6 08 00 00 f0 00 00 00 f0 00 cc 15 91 08 55 43 f9 01 00 00 f0 21 a4 13 91 03 00 00 f0 63 fc 13 91 c2 0f 80 52 00 01 3f d6 08 00 00 f0 00 00 00 f0 00 24 16 91 08 55 43 f9 01
00 00 f0 21 b4 16 91 03 00 00 f0 63 4c 17 91 e2 02 80 52 00 01 3f d6 fd 7b bf a9 08 00 00 f0 fd 03 00 91 08 4d 43 f9 08 3d 00 12 1f 05 00
71 01 04 00 54 88 2e 40 f9 1f 05 00 f1 e0 04 00 54 08 01 40 b2 a8 86 00 f8 88 2a 41 39 1f 0d 00 71 c0 02 00 54 88 02 40 f9 1f 05 00 f1 60 06 00 54 e9 03 28 2a 3f 05 40 f2 40 07 00 54 08 f9
7f 92 0a 00 00 f0 09 05 40 f9 4a 51 43 f9 3f 01 0a eb c1 07 00 54 09 49 80 b9 89 0e 09 8b a9 02 09 cb 29 41 01 d1 69 03 10 b7 29 fd 43 d3 08 41 40 b9 1f 01 09 6b eb 02 00 54 fd 7b c1 a8 3d
00 00 14 08 00 00 f0 00 00 00 f0 00 40 13 91 08 55 43 f9 01 00 00 f0 21 a4 13 91 03 00 00 f0 63 fc 13 91 22 0f 80 52 00 01 3f d6 08 00 00 f0 00 00 00 f0 00 3c 15 91 08 55 43 f9 01 00 00 f0
21 a4 13 91 03 00 00 f0 63 fc 13 91 42 0f 80 52 00 01 3f d6 08 00 00 f0 00 cc 15 91 08 55 43 f9 01 00 00 f0 21 a4 13 91 03 00 00 f0 63 fc 13 91 c2 0f 80 52 00 01 3f d6 08 00 00
f0 00 00 00 f0 00 24 16 91 08 55 43 f9 01 00 00 f0 21 b4 16 91 03 00 00 f0 63 4c 17 91 a2 02 80 52 00 01 3f d6 08 00 00 f0 00 00 00 f0 00 88 18 91 08 55 43 f9 01 00 00 f0 21 fc 18 91 03 00
00 f0 63 88 19 91 22 3f 80 52 00 01 3f d6 08 00 00 f0 00 00 00 f0 00 24 18 91 08 55 43 f9 01 00 00 f0 21 b4 16 91 03 00 00 f0 63 4c 17 91 e2 02 80 52 00 01 3f d6 48 6f 01 58 1f 20 03 d5 88
1e 00 f9 1f 20 03 d5
```

Native machine code

`_SET_IP`

_BINARY_OP_ADD_INT

CHECK_PERIODIC

_JUMP_TO_TOP

arm

Native machine code

_JUMP_TO_TOP

arm

The CPython JIT in 3.14

Native machine code

_BINARY_OP_ADD_INT

_STORE_FAST_1

_CHECK_PERIODIC

_JUMP_TO_TOP

```
a8 82 5f f8 e9 03 28 2a 3f 05 40 f2 20 01 00 54 08 f9 7f 92 09 00 00 d0 08 05 40 f9 29 ed 46 f9 1f 01 09 eb 40 00 00 54 66 06 00 14 0d 00 00 14 fd 7b bf a9 08 00 00 d0 00 00 00 d0 00 80 35
91 08 f1 46 f9 01 00 00 d0 21 f4 35 91 03 00 00 00 63 80 36 91 22 3f 80 52 fd 03 00 91 00 01 3f d6 fd 7b bf a9 a8 82 5f f8 fd 03 00 91 e9 03 28 2a 3f 05 40 f2 c0 01 00 54 08 f9 7f 92 0a 00
00 d0 09 05 40 f9 4a 75 47 f9 3f 01 0a eb 41 02 00 54 08 11 40 f9 1f 01 00 00 f1 6d 00 00 54 fd 7b c1 a8 18 00 00 01 14 fd 7b c1 a8 0c 07 00 14 08 00 00 d0 00 00 00 d0 00 44 39 91 08 79 47 f9 01
00 00 d0 21 48 3a 91 03 00 00 00 63 d4 3a 91 22 3f 80 52 00 01 3f d6 08 00 00 d0 00 00 00 a0 37 91 08 79 47 f9 01 00 00 d0 21 20 38 91 03 00 00 d0 63 94 38 91 42 0f 80 52 00 01 3f d6
1f 20 03 d5 fd 7b bf a9 a8 82 5f f8 fd 03 00 91 e9 03 28 2a 3f 05 40 f2 e0 06 00 54 08 f9 7f 92 0a 00 00 f0 09 05 40 f9 4a 79 40 f9 3f 01 0a eb 61 07 00 54 09 11 40 f9 3f 01 00 f1 4d 08 00
91 62 0f 80 52 00 01 3f d6 08 00 00 00 00 00 00 00 00 00 00 ec 3d 91 08 85 40 f9 01 00 00 d0 21 60 3c 91 03 00 00 d0 63 ac 3c 91 e2 0f 80 52 00 01 3f d6 08 00 00 f0 00 00 00 d0 18 3e 91 08 85
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01 3f d6 08 00 00 f0 00 00 00 f0 00 48 01 91 08 85 40 f9 01 00 00 f0 21 d8 01 91 03 00 00 f0 63 70 02 91 a2 02 80 52 00 01 3f d6 08 00 00 f0 00 00 00 00 00 f0 00 48 03 91 08 85 40 f9 01 00 00 f0
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0c 6a e0 00 00 54 5f 05 00 f1 a0 00 00 54 8a 00 00 36 4b f9 7f 92 6b 01 40 b9 4b 13 f8 36 8b 2a 41 39 93 32 40 f9 8a 32 00 f9 7f 0d 00 71 a0 01 00 54 0a 05 40 f9 5f 01 09 eb c1 05 00 54 09
49 80 b9 89 0e 09 8b a9 02 09 cb 29 41 01 d1 29 0a 10 b7 29 fd 43 d3 08 41 40 b9 1f 01 09 6b ab 09 00 54 88 22 40 f9 a8 0a 00 b5 95 22 00 f9 b3 0b 00 b4 b3 03 00 37 7f 06 00 f1 80 0c 00 54
68 02 40 b9 08 01 00 35 08 00 00 f0 00 00 00 f0 00 20 0b 91 08 55 42 f9 41 53 80 52 e2 03 13 aa 00 01 3f d6 68 0e 40 79 68 0c 10 37 68 02 40 b9 a8 00 f8 37 a8 76 01 58 1f 20 03 d5 00 01 3f
d6 68 02 40 b9 08 05 00 71 68 02 00 b9 a1 00 00 54 08 00 00 f0 e0 03 13 aa 08 5d 42 f9 00 01 3f d6 95 22 40 f9 d5 0b 00 b4 9f 22 00 f9 fd 7b c1 a8 70 00 00 14 08 00 00 f0 00 00 00 f0 00 48
0a 91 08 61 42 f9 01 00 00 f0 21 d8 08 91 03 00 00 f0 63 70 09 91 e2 02 80 52 00 01 3f d6 08 00 00 f0 00 00 00 f0 00 60 04 91 08 61 42 f9 01 00 00 f0 21 c4 04 91 03 00 00 f0 63 04 05 91 22
0f 80 88 67 01 58 1f 20 03 d5 08 89 40 39 48 00 00 37 c0 06 00 14 1f 20 03 d5 fd 7b bf a9 08 00 00 f0 fd 03 00 91 08 4d 43 f9 08 3d 00 12 1f 05 00 71 01 04 00 54 88 2e 40 f9 1f 05 00 f1 e0
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91 08 55 43 f9 01 00 00 f0 21 a4 13 91 03 00 00 f0 63 fc 13 91 22 0f 80 52 00 01 3f d6 08 00 00 f0 00 00 00 f0 00 3c 15 91 08 55 43 f9 01 00 00 f0 21 a4 13 91 03 00 00 f0 63 fc 13 91 42 0f
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08 00 00 f0 00 00 00 00 f0 00 24 18 91 08 55 43 f9 01 00 00 f0 21 b4 16 91 03 00 00 f0 63 4c 17 91 e2 02 80 52 00 01 3f d6 fd 7b bf a9 08 00 00 f0 fd 03 00 91 08 4d 43 f9 08 3d 00 12 1f 05 00
71 01 04 00 54 88 2e 40 f9 1f 05 00 f1 e0 04 00 54 08 01 40 b2 a8 86 00 f8 88 2a 41 39 1f 0d 00 71 c0 02 00 54 88 02 40 f9 1f 05 00 f1 60 06 00 54 e9 03 28 2a 3f 05 40 f2 40 07 00 54 08 f9
7f 92 0a 00 00 f0 09 05 40 f9 4a 51 43 f9 3f 01 0a eb c1 07 00 54 09 49 80 b9 89 0e 09 8b a9 02 09 cb 29 41 01 d1 69 03 10 b7 29 fd 43 d3 08 41 40 b9 1f 01 09 6b eb 02 00 54 fd 7b c1 a8 3d
00 00 14 08 00 00 f0 00 00 00 f0 00 40 13 91 08 55 43 f9 01 00 00 f0 21 a4 13 91 03 00 00 f0 63 fc 13 91 22 0f 80 52 00 01 3f d6 08 00 00 f0 00 00 00 f0 00 3c 15 91 08 55 43 f9 01 00 00 f0
21 a4 13 91 03 00 00 f0 63 fc 13 91 42 0f 80 52 00 01 3f d6 08 00 00 f0 00 00 00 f0 00 cc 15 91 08 55 43 f9 01 00 00 f0 21 a4 13 91 03 00 00 f0 63 fc 13 91 c2 0f 80 52 00 01 3f d6 08 00 00
f0 00 00 00 f0 00 24 16 91 08 55 43 f9 01 00 00 f0 21 b4 16 91 03 00 00 f0 63 4c 17 91 a2 02 80 52 00 01 3f d6 08 00 00 f0 00 00 00 f0 00 88 18 91 08 55 43 f9 01 00 00 f0 21 fc 18 91 03 00
00 f0 63 88 19 91 22 3f 80 52 00 01 3f d6 08 00 00 f0 00 00 00 00 f0 24 18 91 08 55 43 f9 01 00 00 f0 21 b4 16 91 03 00 00 f0 63 4c 17 91 e2 02 80 52 00 01 3f d6 48 6f 01 58 1f 20 03 d5 88
1e 00 f9 1f 20 03 d5 fd 7b bf a9 b3 02 5f f8 fd 03 00 91 e8 03 33 2a 1f 05 40 f2 e0 13 00 54 b7 82 5f f8 e8 03 37 2a 1f 05 40 f2 60 13 00 54 60 fa 7f 92 08 00 00 f0 09 04 40 f9 08 0d 47 f9
3f 01 08 eb 21 15 00 54 e1 fa 7f 92 29 04 40 f9 3f 01 08 eb e1 15 00 54 88 22 40 f9 e8 16 00 b5 95 22 00 09 f9 68 89 01 58 1f 20 03 d5 00 01 3f d6 98 22 40 f9 78 17 00 b4 ff 06 00 f1 9f 22 00
f9 00 12 00 54 f5 03 00 aa 37 04 00 37 e8 02 40 b9 08 01 00 35 08 00 00 f0 00 00 00 00 f0 e0 00 28 2d 91 08 15 47 f9 a1 51 80 52 e2 03 17 aa 00 01 3f d6 e8 0e 40 79 a8 16 10 37 e8 02 40 b9 a8 00
f8 37 08 87 01 58 1f 20 03 d5 00 01 3f d6 e8 02 40 b9 08 05 00 71 e8 02 00 b9 a1 01 00 54 69 86 01 58 1f 20 03 d5 28 c1 73 f9 a8 00 00 b4 22 c5 73 f9 e0 03 17 aa 21 00 80 52 00 01 3f d6 08
00 00 f0 e0 03 17 aa 08 21 47 f9 00 01 3f d6 7f 06 00 f1 80 0d 00 54 b3 02 00 37 68 02 40 b9 08 01 00 35 08 00 00 f0 00 00 00 00 f0 00 28 2d 91 08 15 47 f9 a1 51 80 52 e2 03 13 aa 00 01 3f d6
68 0e 40 79 48 12 10 37 68 02 40 b9 a8 00 f8 37 a8 82 01 58 1f 20 03 d5 00 01 3f d6 68 02 40 b9 08 05 00 71 68 02 00 00 00 00 b9 a0 04 00 54 35 06 00 b4 a8 0e 40 79 08 01 40 92 08 01 15 aa 88 00 00
37 a9 fa 7f 92 29 0d 40 79 69 16 10 37 08 03 1f f8 15 23 00 d1 88 2a 41 39 1f 0d 00 71 c0 02 00 54 88 02 40 f9 1f 05 00 f1 a0 11 00 54 e9 03 28 2a 3f 05 40 f2 40 07 00 54 08 f9 7f 92 0a 00
00 f0 09 05 40 f9 4a 25 47 f9 3f 01 0a eb c1 11 00 54 09 49 80 b9 89 0e 09 8b a9 02 09 cb 29 41 01 d1 a9 0e 10 b7 29 fd 43 d3 08 41 40 b9 1f 01 09 6b 2b 0e 00 54 fd 7b c1 a8 a1 00 00 14 89
7d 01 58 1f 20 03 d5 28 c1 73 f9 a8 00 00 b4 22 c5 73 f9 e0 03 13 aa 21 00 80 52 00 01 3f d6 08 00 00 f0 e0 03 13 aa 08 21 47 f9 00 01 3f d6 35 fa ff b5 88 2a 41 39 15 43 00 d1 1f 0d 00 71
c0 02 00 54 88 02 40 f9 1f 05 00 f1 a0 0c 00 54 e9 03 28 2a 3f 05 40 f2 40 02 00 54 08 f9 88 6a 01 58 1f 20 03 d5 08 89 40 39 48 00 00 37 04 05 00 14 1f 20 03 d5 48 6f 01 58 1f 20 03 d5 88
1e 00 f9 1f 20 03 d5 a8 02 5f f8 e9 03 28 2a 3f 05 40 f2 20 01 00 54 08 00 00 d0 08 05 40 f9 29 71 47 f9 1f 01 09 eb 40 00 00 54 34 05 00 14 0d 00 00 14 fd 7b bf a9 08 00 00 d0
00 00 00 d0 00 a0 39 91 08 75 47 f9 01 00 00 d0 21 14 3a 91 03 00 00 d0 63 a0 3a 91 22 3f 80 52 fd 03 00 91 00 01 3f d6
```

The CPython JIT in 3.14

Native machine code

_STORE_FAST_1

_CHECK_PERIODIC

_JUMP_TO_TOP

```
a8 82 5f f8 e9 03 28 2a 3f 05 40 f2 20 01 00 54 08 f9 7f 92 09 00 00 d0 08 05 40 f9 29 ed 46 f9 1f 01 09 eb 40 00 00 54 66 06 00 14 0d 00 00 14 fd 7b bf a9 08 00 00 d0 00 00 00 d0 00 80 35
91 08 f1 46 f9 01 00 00 d0 21 f4 35 91 03 00 00 00 d0 63 80 36 91 22 3f 80 52 fd 03 00 91 00 01 3f d6 fd 7b bf a9 a8 82 5f f8 fd 03 00 91 e9 03 28 2a 3f 05 40 f2 e0 06 00 54 08 f9 7f 92 09 00 00 54 66 06 00 14 0d 00 00 14 fd 7b bf a9 08 00 00 d0 00 00 00 d0 00 80 35
00 d0 09 05 40 f9 4a 75 47 f9 3f 01 0a eb 41 02 00 54 08 11 40 f9 1f 01 00 00 f1 6d 00 00 54 fd 7b c1 a8 18 00 00 01 14 fd 7b c1 a8 0c 07 00 14 08 00 00 d0 00 00 00 d0 00 d4 39 91 08 79 47 f9 01
00 00 d0 21 48 3a 91 03 00 00 d0 63 d4 3a 91 22 3f 80 52 00 01 3f d6 08 00 f0 d0 00 a0 37 91 08 79 47 f9 01 00 00 d0 21 20 38 91 03 00 00 d0 63 94 38 91 42 0f 80 52 00 01 3f d6
1f 20 03 d5 fd 7b bf a9 a8 82 5f f8 fd 03 00 91 e9 03 28 2a 3f 05 40 f2 e0 06 00 54 08 f9 7f 92 09 00 00 54 66 06 00 14 0d 00 00 14 fd 7b bf a9 08 00 00 d0 00 00 00 d0 00 80 35
91 62 0f 80 52 00 01 3f d6 08 00 00 f0 00 00 00 00 d0 00 ec 3d 91 08 85 40 f9 01 00 00 d0 21 60 3c 91 03 00 00 d0 63 ac 3c 91 e2 0f 80 52 00 01 3f d6 08 00 00 f0 00 00 00 d0 00 18 3e 91 08 85
40 f9 01 00 00 d0 21 60 3c 91 03 00 00 d0 63 ac 3c 91 e2 0f 80 52 00 01 3f d6 08 00 00 f0 00 00 00 d0 63 3c 00 f0 91 03 00 00 f0 63 3c 00 f0 91 62 3b 80 52 00
01 3f d6 08 00 00 f0 00 00 00 f0 00 48 01 91 08 85 40 f9 01 00 00 f0 21 d8 01 91 03 00 00 f0 63 70 02 91 e2 02 80 52 00 01 3f d6 08 00 00 f0 00 00 00 d0 00 18 3e 91 08 85
21 d8 01 91 03 00 00 f0 63 70 02 91 e2 02 80 52 00 01 3f d6 88 62 01 58 1f 20 03 d5 88 1e 00 f9 1f 20 03 d5 fd 7b bf a9 08 00 00 f0 fd 03 00 91 08 4d 42 f9 08 3d 00 12 1f 09 00 71 e1 0a 00
54 88 02 40 f9 1f 05 00 f1 c0 0b 00 54 e9 03 28 2a 3f 05 40 f2 a0 0c 00 54 08 f9 7f 92 09 00 00 f0 0a 05 40 f9 29 51 42 f9 5f 01 09 eb 21 08 00 54 0b c1 40 39 0c 14 80 52 aa 8e 5f f8 7f 01
0c 6a e0 00 00 54 5f 05 00 f1 a0 00 00 54 8a 00 00 36 4b f9 7f 92 6b 01 40 b9 4b 13 f8 36 8b 2a 41 39 93 32 40 f9 8a 32 00 f9 7f 0d 00 71 a0 01 00 54 0a 05 40 f9 5f 01 09 eb c1 05 00 54 09
49 80 b9 89 0e 09 8b a9 02 09 cb 29 41 01 d1 29 0a 10 b7 29 fd 43 d3 08 41 40 b9 1f 01 09 6b ab 09 00 54 88 22 40 f9 a8 0a 00 b5 95 22 00 f9 b3 0b 00 b4 b3 03 00 37 7f 06 00 f1 80 0c 00 54
68 02 40 b9 08 01 00 35 08 00 00 f0 00 00 00 f0 00 20 0b 91 08 55 42 f9 41 53 80 52 e2 03 13 aa 00 01 3f d6 68 0e 40 79 68 0c 10 37 68 02 40 b9 a8 00 f8 37 a8 76 01 58 1f 20 03 d5 00 01 3f
d6 68 02 40 b9 08 05 00 71 68 02 00 b9 a1 00 00 54 08 00 00 f0 e0 03 13 aa 08 5d 42 f9 00 01 3f d6 95 22 40 f9 d5 0b 00 b4 9f 22 00 f9 fd 7b c1 a8 70 00 00 14 08 00 00 f0 00 00 00 f0 00 48
0a 91 08 61 42 f9 01 00 00 f0 21 d8 08 91 03 00 00 f0 63 70 09 91 e2 02 80 52 00 01 3f d6 08 00 00 f0 00 00 00 f0 00 60 04 91 08 61 42 f9 01 00 00 f0 21 c4 04 91 03 00 00 f0 63 04 05 91 22
0f 80 88 67 01 58 1f 20 03 d5 08 89 40 39 48 00 00 37 c0 06 00 14 1f 20 03 d5 fd 7b bf a9 08 00 00 f0 fd 03 00 91 08 4d 43 f9 08 3d 00 12 1f 05 00 71 01 04 00 54 88 2e 40 f9 1f 05 00 f1 e0
04 00 54 08 01 40 b2 a8 86 00 f8 88 2a 41 39 1f 0d 00 71 c0 02 00 54 88 02 40 f9 1f 05 00 f1 63 06 06 00 54 e9 03 28 2a 3f 05 40 f2 a0 07 00 54 08 f9 7f 92 0a 00 00 f0 09 05 40 f9 4a 51 43 09
3f 01 0a eb c1 07 00 54 09 49 80 b9 89 0e 09 8b a9 02 09 cb 29 41 01 d1 69 03 10 b7 29 fd 43 d3 08 41 40 b9 1f 01 09 6b ab 09 00 54 88 22 40 f9 a8 0a 00 b5 95 22 00 f9 b3 0b 00 b4 b3 03 00 37 7f 06 00 f1 80 0c 00 54
91 08 55 43 f9 01 00 00 f0 21 a4 13 91 03 00 00 00 63 fc 13 91 22 0f 80 52 00 01 3f d6 08 00 00 f0 00 00 00 f0 00 c3 15 91 08 55 43 f9 01 00 00 f0 21 a4 13 91 03 00 00 f0 63 fc 13 91 42 0f
80 52 00 01 3f d6 08 00 00 f0 00 00 00 cc 15 91 08 55 43 f9 01 00 00 00 21 a4 13 91 03 00 00 f0 63 fc 13 91 c2 0f 80 52 00 01 3f d6 08 00 00 f0 00 24 16 91 08 55 43 f9 01 00 00 f0 63 fc 13 91 42 0f
00 00 f0 21 b4 16 91 03 00 00 f0 63 4c 17 91 a2 02 80 52 00 01 3f d6 08 00 00 f0 00 00 00 f0 88 18 91 08 55 43 f9 01 00 00 f0 21 c0 02 00 54 88 02 40 f9 1f 05 00 f1 60 06 00 54 e9 03 28 2a 3f 05 40 f2 a0 07 00 54 08 f9
08 00 00 f0 00 00 00 f0 00 24 18 91 08 55 43 f9 01 00 00 f0 21 b4 16 91 03 00 00 f0 63 4c 17 91 e2 02 80 52 00 01 3f d6 08 00 00 f0 00 00 00 f0 00 24 16 91 08 55 43 f9 01 00 00 f0 63 fc 13 91 42 0f
71 01 04 00 54 88 2e 40 f9 1f 05 00 f1 e0 04 00 54 08 01 40 b2 a8 86 00 f8 88 2a 41 39 1f 0d 00 71 c0 02 00 54 88 02 40 f9 1f 05 00 f1 60 06 00 54 e9 03 28 2a 3f 05 40 f2 a0 07 00 54 08 f9
7f 92 0a 00 00 f0 09 05 40 f9 4a 51 43 f9 3f 01 0a eb c1 07 00 54 09 49 80 b9 89 0e 09 8b a9 02 09 cb 29 41 01 d1 69 03 10 b7 29 fd 43 d3 08 41 40 b9 1f 01 09 6b ab 09 00 54 88 22 40 f9 a8 0a 00 b5 95 22 00 f9 b3 0b 00 b4 b3 03 00 37 7f 06 00 f1 80 0c 00 54
00 00 14 08 00 00 f0 00 00 00 f0 00 40 13 91 08 55 43 f9 01 00 00 f0 21 a4 13 91 03 00 00 f0 63 fc 13 91 22 0f 80 52 00 01 3f d6 08 00 00 f0 00 00 00 f0 00 24 16 91 08 55 43 f9 01 00 00 f0 63 fc 13 91 42 0f
21 a4 13 91 03 00 00 f0 63 fc 13 91 42 0f 80 52 00 01 3f d6 08 00 00 f0 00 cc 15 91 08 55 43 f9 01 00 00 f0 21 a4 13 91 03 00 00 f0 63 fc 13 91 c2 0f 80 52 00 01 3f d6 08 00 00 f0 00 24 16 91 08 55 43 f9 01 00 00 f0 63 fc 13 91 42 0f
f0 00 00 00 f0 00 24 16 91 08 55 43 f9 01 00 00 f0 21 b4 16 91 03 00 00 f0 63 4c 17 91 a2 02 80 52 00 01 3f d6 08 00 00 f0 00 00 00 f0 00 00 88 18 91 08 55 43 f9 01 00 00 f0 21 fc 18 91 03 00
00 f0 63 88 19 91 22 3f 80 52 00 01 3f d6 08 00 00 f0 00 24 18 91 08 55 43 f9 01 00 00 f0 21 b4 16 91 03 00 00 f0 63 4c 17 91 e2 02 80 52 00 01 3f d6 48 6f 01 58 1f 20 03 d5 88
1e 00 f9 1f 20 03 d5 fd 7b bf a9 b3 02 5f f8 fd 03 00 91 e8 03 33 2a 1f 05 40 f2 e0 13 00 54 b7 82 5f f8 e8 03 37 2a 1f 05 40 f2 60 13 00 54 60 fa 7f 92 08 00 00 f0 09 04 40 f9 08 0d 47 f9
3f 01 08 eb 21 15 00 54 e1 fa 7f 92 29 04 40 f9 3f 01 08 eb e1 15 00 54 88 22 40 f9 e8 16 00 b5 95 22 00 f9 68 89 01 58 1f 20 03 d5 00 01 3f d6 98 22 40 f9 78 17 00 b4 ff 06 00 f1 9f 22 00
f9 00 12 00 54 f5 03 00 aa 37 04 00 37 e8 02 40 b9 08 01 00 35 08 00 00 f0 00 00 28 2d 91 08 15 47 f9 a1 51 80 52 e2 03 17 aa 00 01 3f d6 e8 0e 40 79 a8 16 10 37 e8 02 40 b9 a8 00
f8 37 08 87 01 58 1f 20 03 d5 00 01 3f d6 e8 02 40 b9 08 05 00 71 e8 02 00 b9 a1 01 00 54 69 86 01 58 1f 20 03 d5 28 c1 73 f9 a8 00 00 b4 22 c5 73 f9 e0 03 17 aa 21 00 80 52 00 01 3f d6 08
00 00 f0 e0 03 17 aa 08 21 47 f9 00 01 3f d6 7f 06 00 f1 80 0d 00 54 b3 02 00 37 68 02 40 b9 08 01 00 35 08 00 00 f0 00 00 00 f0 00 28 2d 91 08 15 47 f9 a1 51 80 52 e2 03 13 aa 00 01 3f d6
68 0e 40 79 48 12 10 37 68 02 40 b9 a8 00 00 f8 37 a8 82 01 58 1f 20 03 d5 00 01 3f d6 68 02 40 b9 08 05 00 71 68 02 40 00 b9 a0 04 00 54 35 06 00 b4 a8 0e 40 79 08 01 40 92 08 01 15 aa 88 00 00
37 a9 fa 7f 92 29 0d 40 79 69 16 10 37 08 03 1f f8 15 23 00 d1 88 2a 41 39 1f 0d 00 71 c0 02 00 54 88 02 40 f9 1f 05 00 f1 a0 11 00 54 e9 03 28 2a 3f 05 40 f2 a0 07 00 54 08 f9 7f 92 0a 00
00 f0 09 05 40 f9 4a 25 47 f9 3f 01 0a eb c1 11 00 54 09 49 80 b9 89 0e 09 8b a9 02 09 cb 29 41 01 d1 a9 0e 10 b7 29 fd 43 d3 08 41 40 b9 1f 01 09 6b 2b 0e 00 54 fd 7b c1 a8 a1 00 00 14 89
7d 01 58 1f 20 03 d5 28 c1 73 f9 a8 00 00 b4 22 c5 73 f9 e0 03 13 aa 21 00 80 52 00 01 3f d6 08 00 00 f0 e0 03 13 aa 08 21 47 f9 00 01 3f d6 35 fa ff b5 88 2a 41 39 15 43 00 d1 1f 0d 00 71
c0 02 00 54 88 02 40 f9 1f 05 00 f1 a0 0c 00 54 e9 03 28 2a 3f 05 40 f2 a0 02 00 54 08 f9 88 6a 01 58 1f 20 03 d5 08 89 40 39 48 00 00 37 04 05 00 14 1f 20 03 d5 48 6f 01 58 1f 20 03 d5 88
1e 00 f9 1f 20 03 d5 a8 02 5f f8 e9 03 28 2a 3f 05 40 f2 20 01 00 54 08 f9 7f 92 09 00 00 d0 08 05 40 f9 29 71 47 f9 1f 01 09 eb 40 00 00 54 34 05 00 14 0d 00 00 14 fd 7b bf a9 08 00 00 d0
00 00 00 d0 00 a0 39 91 08 75 47 f9 01 00 00 d0 21 14 3a 91 03 00 00 d0 63 a0 3a 91 22 3f 80 52 fd 03 00 91 00 01 3f d6 fd 7b bf a9 b3 02 5f f8 fd 03 00 91 e8 03 33 2a 1f 05 40 f2 e0 13 00
54 b7 82 5f f8 e8 03 37 2a 1f 05 40 f2 60 13 00 54 60 fa 7f 92 08 00 00 f0 09 04 40 f9 08 45 41 f9 3f 01 08 eb 21 15 00 54 e1 fa 7f 92 29 04 40 f9 3f 01 08 eb e1 15 00 54 88 22 40 f9 e8 16
00 b5 95 22 00 f9 28 85 01 58 1f 20 03 d5 00 01 3f d6 98 22 40 f9 78 17 00 b4 ff 06 00 f0 19 22 00 f9 00 12 00 54 f5 03 00 aa 37 04 00 37 e8 02 40 b9 08 01 00 35 08 00 00 f0 00 00 00 d0 00
f4 3e 91 08 4d 41 f9 a1 51 80 52 e2 03 17 aa 00 01 3f d6 e8 0e 40 79 a8 16 10 37 e8 02 40 b9 a8 00 f8 37 c8 82 01 58 1f 20 03 d5 00 01 3f d6 e8 02 40 b9 08 05 00 71 e8 02 00 b9 a1 01 00 54
29 82 01 58 1f 20 03 d5 28 c1 73 f9 a8 00 00 b4 22 c5 73 f9 e0 03 17 aa 21 00 80 52 00 01 3f d6 08 00 00 f0 e0 03 17 aa 08 59 41 f9 00 01 3f d6 7f 06 00 f1 80 0d 00 54 b3 02 00 37 68 02 40
b9 08 01 00 35 08 00 00 f0 00 00 d0 00 00 f4 3e 91 08 4d 41 f9 a1 51 80 52 e2 03 13 aa 00 01 3f d6 68 0e 40 79 48 12 10 37 68 02 40 b9 a8 00 f8 37 68 7e 01 58 1f 20 03 d5 00 01 3f d6 68 02
40 b9 08 05 00 71 68 02 00 b9 a0 04 00 54 35 06 00 b4 a8 0e 40 79 08 01 40 92 08 01 15 aa 88 00 00 37 a9 fa 7f 92 29 0d 40 79 69 16 10 37 08 03 1f f8 15 23 00 d1 88 2a 41 39 1f 0d 00 71 c0
02 00 54 88 02 40 f9 1f 05 00 f1 a0 11 00 54 e9 03 28 2a 3f 05 40 f2 a0 02 00 54 08 f9 7f 92 0a 00 00 f0 09 05 40 f9 4a 5d 41 f9 3f 01 0a eb c1 11 00 54 09 49 80 b9 89 0e 09 8b a9 02 09 cb
29 41 01 d1 a9 0e 10 b7 29 fd 43 d3 08 41 40 b9 1f 01 09 6b 2b 0e 00 54 fd 7b c1 a8 a1 00 00 14 49 79 01 58 1f 20 03 d5 28 c1 73 f9 a8 00 00 b4 22 c5 73 f9 e0 03 13 aa 21 00 80 52 00 01 3f
d6 08 00 00 f0 e0 03 13 aa 08 59 41 f9 00 01 3f d6 35 fa ff b5 88 2a 41 39 15 43 00 d1 1f 0d 00 71 c0 02 00 54 88 02 40 f9 1f 05 00 f1 a0 0c 00 54 e9 03 28 2a 3f 05 40 f2 40 02 00 54 08 f9
7f 92
```


Native machine code

91	82	f1	f8	e9	03	28	2a	3f	05	f0	d2	20	01	00	5d	00	f7	92	09	00	d0	08	05	40	f9	29	ed	46	f9	f1	09	eb	40	00	00	54	f8	06	03	14	d0	00	00	14	fd	7b	bf	a9	08	00	d0	00	00	00	00	00	00	00	80	35				
00	08	f1	46	f9	01	00	00	21	f4	35	91	03	00	00	d0	63	80	36	91	22	3f	d0	80	52	fd	03	00	91	00	01	3f	fe	fd	7b	bf	a9	a8	82	f5	f8	fd	00	00	91	e9	03	28	2a	3f	b5	a0	f2	c0	01	00	54	08	f9	7f	92	0a	00		
00	00	09	05	40	f9	4a	75	47	f9	3f	01	0a	eb	41	02	00	54	08	11	40	f9	1f	01	00	f1	6d	00	00	54	fd	7b	c1	a8	18	00	00	14	fd	7b	c1	a8	0c	07	00	14	08	00	00	d4	39	91	08	79	47	f9	01								
00	00	d0	21	48	3a	91	03	00	00	d0	63	d4	3a	91	22	3f	80	52	00	01	3f	d6	08	00	00	d0	00	00	d0	00	00	00	00	37	91	08	79	47	f9	01	00	d0	21	20	38	91	03	00	d0	00	63	94	38	91	00	01	3f	d6	00					
1f	20	03	d5	fd	7b	bf	a9	a8	82	f5	f8	fd	03	00	91	e9	03	28	2a	3f	05	40	f2	e0	06	00	54	08	f9	7f	92	0a	00	00	f0	f9	05	40	f9	4a	79	40	f9	3f	01	0a	eb	61	07	00	54	09	11	40	f9	3f	01	00	f1	4d	08	00		
91	62	f1	00	52	00	01	3f	d6	08	00	00	f0	00	00	d0	ec	3d	91	08	85	40	f9	01	00	00	00	00	d0	21	60	3c	91	03	00	d0	00	63	ac	3c	91	e2	0f	80	52	00	01	3f	d6	08	00	f0	00	00	00	d0	00	18	3e	91	08	85			
40	f9	01	00	00	d0	21	60	3c	91	03	00	d0	63	ac	3c	91	42	11	80	52	00	01	3f	d6	08	00	f0	f0	00	00	00	00	00	dc	00	91	85	40	f9	01	00	00	d0	21	e4	3e	91	03	00	f0	63	3c	00	91	62	3b	80	52	00					
01	3f	d6	08	00	00	f0	00	00	f0	00	f0	00	00	00	f0	00	01	91	08	85	40	f9	01	00	00	f0	21	d8	01	91	03	00	f0	00	63	70	02	91	a2	02	80	52	00	01	3f	d6	08	00	f0	00	00	00	d0	00	48	03	91	08	85	40	f9	01	00	f0
21	48	01	91	03	00	00	f0	63	70	02	91	e2	02	80	52	00	01	3f	d6	88	62	01	58	1f	20	03	d5	fd	7b	bf	a9	08	00	f0	00	d0	00	f0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	f0	00		
54	88	02	40	f9	1f	05	00	f1	c0	0b	00	54	e9	03	28	2a	3f	05	40	f2	a0	0c	00	54	08	f9	7f	92	0a	00	f0	0a	05	40	f9	29	51	09	e2	f9	01	09	eb	21	08																			

Native machine code

[illegible]

The CPython JIT in 3.14

Copy-and-patch: an example

```
case _SET_IP: {  
    PyObject *instr_ptr = (PyObject *)CURRENT_OPERAND0();  
    frame->instr_ptr = (_Py_CODEUNIT *)instr_ptr;  
    break;  
}
```

The CPython JIT in 3.14

Copy-and-patch: an example

```
_Py_CODEUNIT *_JIT_ENTRY(_PyInterpreterFrame *frame, _PyStackRef *stack_pointer, PyThreadState *tstate)
{
    PATCH_VALUE(_PyExecutorObject *, current_executor, _JIT_EXECUTOR)
    int oparg;
    int uopcode = _JIT_OPCODE;
    _Py_CODEUNIT *next_instr;
    PATCH_VALUE(uint16_t, _oparg, _JIT_OPARG)
    PATCH_VALUE(uint64_t, _operand0, _JIT_OPERAND0)
    PATCH_VALUE(uint64_t, _operand1, _JIT_OPERAND1)
    PATCH_VALUE(uint32_t, _target, _JIT_TARGET)
    switch (uopcode) {
        case _SET_IP: {
            PyObject *instr_ptr = (PyObject *)CURRENT_OPERAND0();
            frame->instr_ptr = (_Py_CODEUNIT *)instr_ptr;
            break;
        }
    }
    PATCH_VALUE(jit_func_preserve_none, jump, _JIT_CONTINUE);
    return jump(frame, stack_pointer, tstate);
}
```

The CPython JIT in 3.14

Copy-and-patch: an example

```
_Py_CODEUNIT *_JIT_ENTRY(_PyInterpreterFrame *frame, _PyStackRef *stack_pointer, PyThreadState *tstate)
{
    PATCH_VALUE(_PyExecutorObject *, current_executor, _JIT_EXECUTOR)
    int oparg;
    int uopcode = _JIT_OPCODE;
    _Py_CODEUNIT *next_instr;
    PATCH_VALUE(uint16_t, _oparg, _JIT_OPARG)
    PATCH_VALUE(uint64_t, _operand0, _JIT_OPERAND0)
    PATCH_VALUE(uint64_t, _operand1, _JIT_OPERAND1)
    PATCH_VALUE(uint32_t, _target, _JIT_TARGET)
    switch (uopcode) {
        case _SET_IP: {
            PyObject *instr_ptr = (PyObject *)CURRENT_OPERAND0();
            frame->instr_ptr = (_Py_CODEUNIT *)instr_ptr;
            break;
        }
    }
    PATCH_VALUE(jit_func_preserve_none, jump, _JIT_CONTINUE);
    return jump(frame, stack_pointer, tstate);
}
```

The CPython JIT in 3.14

Copy-and-patch: an example

```
0000000000000000 <_JIT_ENTRY>:
    0: 90000008      adrp      x8, 0x0 <_JIT_ENTRY>
           0000000000000000:  R_AARCH64_ADR_GOT_PAGE      _JIT_OPERAND0
    4: f9400108      ldr      x8, [x8]
           0000000000000004:  R_AARCH64_LD64_GOT_LO12_NC  _JIT_OPERAND0
    8: f9001e88      str      x8, [x20, #0x38]
   c: 14000000      b        0xc <_JIT_ENTRY+0xc>
           000000000000000c:  R_AARCH64_JUMP26           _JIT_CONTINUE
```

The CPython JIT in 3.14

Copy-and-patch: an example

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    const unsigned char code_body[16] = {
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        0x88, 0x1e, 0x00, 0xf9, 0x1f, 0x20, 0x03, 0xd5,
    };
    // 0: OPERAND0
    patch_64(data + 0x0, instruction->operand0);
    memcpy(code, code_body, sizeof(code_body));
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The CPython JIT in 3.14

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```

The CPython JIT in 3.14

Copy-and-patch: an example

```
_SET_IP (before patching):  
0x08, 0x00, 0x00, 0x90,      // adrp x8, _JIT_OPERAND0  
0x08, 0x01, 0x40, 0xf9,      // ldr x8, [x8]  
0x88, 0x1e, 0x00, 0xf9,      // str x8, [x20, #0x38]  
0x1f, 0x20, 0x03, 0xd5,      // nop (branch was removed)
```

```
patch_64(data + 0x0, instruction->operand0);  
memcpy(code, code_body, sizeof(code_body));  
patch_aarch64_33rx(code + 0x0, (uintptr_t)data);
```

```
_SET_IP (after patching):  
0xc8, 0x65, 0x01, 0x58,      // ldr x8, <relaxed literal address>  
0x1f, 0x20, 0x03, 0xd5,      // nop  
0x88, 0x1e, 0x00, 0xf9,      // str x8, [x20, #0x38]  
0x1f, 0x20, 0x03, 0xd5,      // nop (branch was removed)
```

The CPython JIT in 3.14

Still experimental

Performance

- From 10% slower to 20% faster, depending on workload

Limitations

- Not compatible with free threading yet
- It requires LLVM to build the JIT stencils

Lack of support for native debuggers (E.g.: gdb, perf)

- Unable to unwind through JIT frames

Platform availability

- Architectures: AArch64, x86_64, i686 (Windows only)
- Operating systems: macOS (clang), Windows (MSVC), Linux (GCC)

Looking forward

Looking Forward

Python 3.15 (pre-alpha)



Supporting stack unwinding in the JIT compiler

[GH-126910](#)



Make the JIT thread-safe

[GH-133171](#)

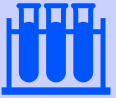


Lots more performance work happening behind the scenes!

Currently we have 40% faster on some benchmarks

Looking Forward

How can you help?



Evaluate, test and debug it

⚠ But not in production! ⚠



Official macOS and Windows release
include the JIT compiler

```
PYTHON_JIT=1
```



Linux

```
./configure --enable-experimental-jit
```



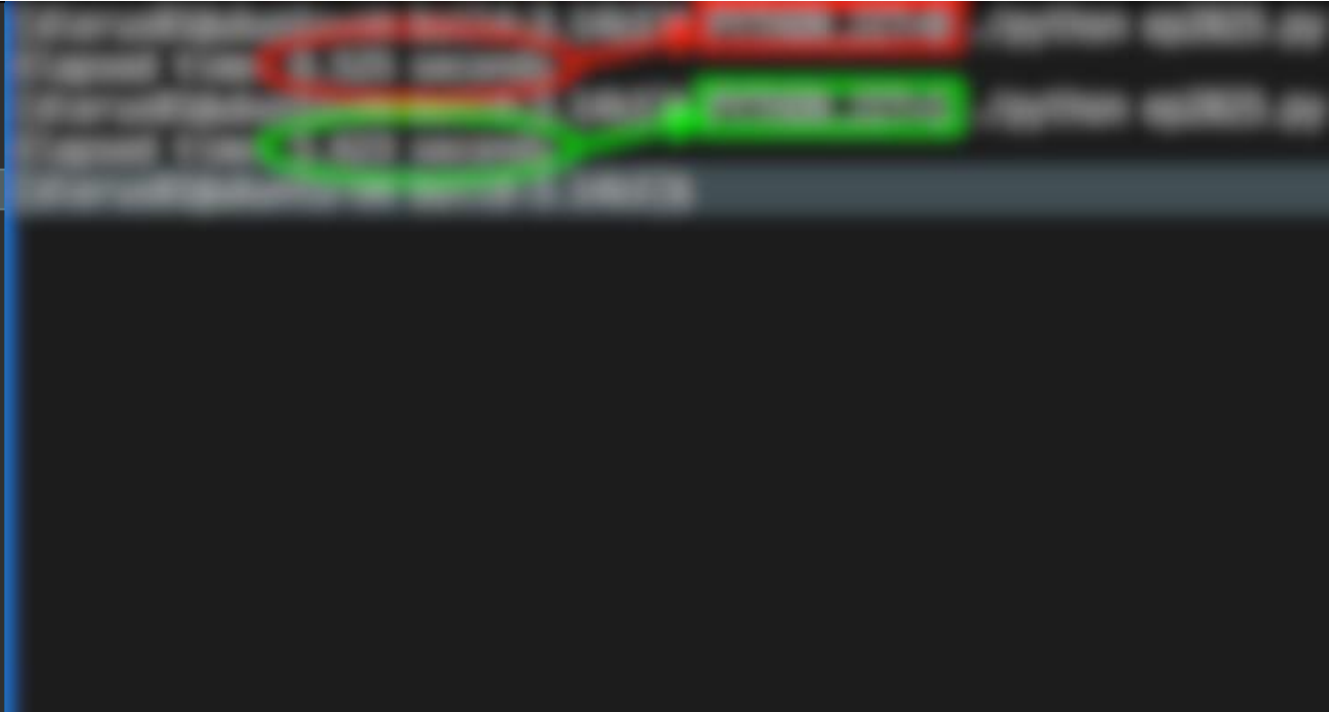
Utilities for helping with the debug
experience

```
_jit.is_available()  
_jit.is_enabled()  
_jit.is_active()
```

Demo

No JIT vs JIT

```
ep2025.py > sum_squares
1 import time
2
3 def sum_squares(n):
4     total = 0
5     for i in range(n):
6         total += i * i
7     return total
8
9 N = 100_000
10
11 start = time.perf_counter()
12 for _ in range(500):
13     sum_squares(N)
14 end = time.perf_counter()
15
16 print(f"Elapsed time: {end - start:.3f} seconds")
```



References



[PEP 659](#)

Specializing Adaptive Interpreter



[PEP 744](#)

JIT Compilation



[Copy-and-patch compilation](#)



[Tools/jit/README.md](#)



Brandt's previous talks

PyCon US 2024: [Building a JIT compiler for CPython](#)

PyCon US 2025: [What they don't tell you about building a JIT compiler for CPython](#)



arm

Merci

Danke

Gracias

Grazie

谢谢

ありがとう

Asante

Thank You

감사합니다

धन्यवाद

Kiitos

شكراً

ধন্যবাদ

תודה

ధన్యవాదములు

Köszönöm